

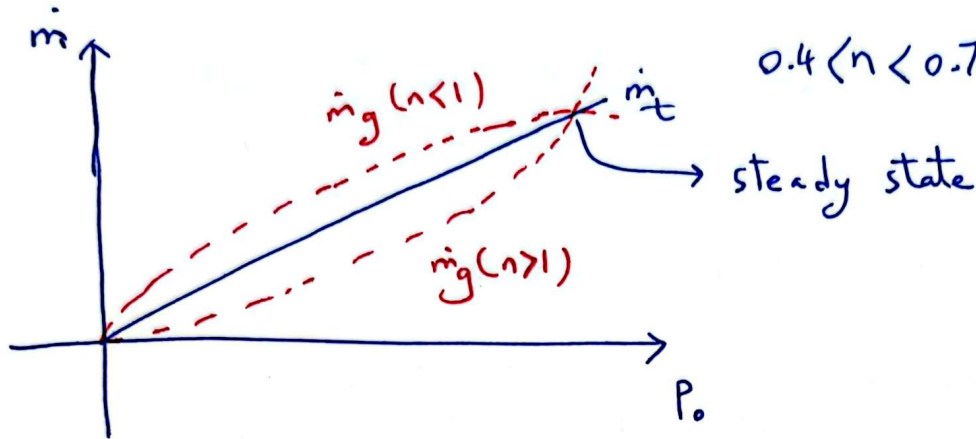
## Burning stability

we know that  $r = a P_o^n$

$n > 1$  : explosive behavior

$n < 1$  : stable "

$0.4 < n < 0.7$  (typical range)



we define two sensitivities for solid rockets

① The sensitivity of burning rate to initial propellant temperature is defined as:

$$\pi_r = \frac{1}{r} \left( \frac{\partial r}{\partial T_p} \right)_{P_o} = \left[ \frac{\partial}{\partial T_p} \ln(r) \right]_{P_o} \quad (12.30)$$

$$r = a P_o^n, \quad \ln r = \ln a + n \ln P_o$$

$$\Rightarrow \pi_r = \left[ \frac{\partial \ln r}{\partial T_p} \right]_{P_o} = \frac{d \ln a}{d T_p} + 0$$

$$a = \frac{A}{T_i - T_p} \Rightarrow \ln a = \ln A - \ln (T_i - T_p)$$

$$\Rightarrow \frac{d \ln a}{d T_p} = + \frac{1}{T_i - T_p} \quad \text{or} \quad \frac{1}{a} \frac{da}{d T_p} = \frac{1}{T_i - T_p}$$

$$\Rightarrow \pi_r = \frac{1}{T_i - T_p} \quad (12.31)$$

cte  $\leftarrow T_i - T_p \rightarrow$  initial temp. (p-1)

Another expression for (12.30)

$$\pi_r = \frac{1}{r} \frac{\partial r}{\partial T_p} \quad \left\{ \begin{array}{l} r = \frac{r_1 + r_2}{2} \text{ (average } r) \\ \frac{\partial r}{\partial T_p} = \frac{r_2 - r_1}{T_{p_2} - T_{p_1}} \end{array} \right.$$

$$\pi_r = \frac{2}{r_1 + r_2} \frac{r_2 - r_1}{T_{p_2} - T_{p_1}} \quad \text{second expression for } \pi_r$$

$$0.002 \text{ K}^{-1} < \pi_r < 0.008 \text{ K}^{-1} \text{ (typical } \pi_r \text{ range)}$$

(2) The sensitivity of  $p_o$  to initial grain temperature is defined as:

$$\pi_p = \frac{1}{p_o} \left( \frac{\partial p_o}{\partial T_p} \right)_{\frac{A_b}{A_t}, T_o} = \left[ \frac{\partial}{\partial T_p} \ln p_o \right]_{\frac{A_b}{A_t}, T_o}$$

From  $p_o = \left( \frac{A_b}{A_t} a f_p c^x \right)^{\frac{1}{1-n}}$ , and noting that only "a" depends on initial temperature,

$$\left[ \frac{\partial \ln p_o}{\partial T_p} \right]_{\frac{A_b}{A_t}, T_o} = \frac{1}{1-n} \frac{d \ln a}{d T_p} = \frac{1}{1-n} \pi_r \quad (12.32)$$

$$\pi_p = \frac{1}{1-n} \pi_r = \frac{1}{1-n} \frac{1}{T_1 - T_p}$$

(P-2)

To reduce the sensitivity of  $p_0$  to initial grain temperature, the propellant should have high  $T_i$  and lower  $n$ .

Fig. 12.19 shows the impact of the initial grain temperature on  $P_0$ .

Axial pressure variation: (x-direction)



How  $p$  varies with  $x$ ?

$$P = \frac{P_i}{2} \left[ 1 + \sqrt{1 - 4RT_i \left( \frac{s_p c r x}{P_i A} \right)^2} \right]$$

initial pressure
temp.
Circumference of the grain
location

Example: A solid rocket missile burns at the following rates at an initial temperature of  $-10^\circ\text{C}$  at Alaska, border with Russia.

$r = 3 \frac{\text{mm}}{\text{s}}$  at  $20 \text{ atm}$  and  $r = 8 \frac{\text{mm}}{\text{s}}$  at  $70 \text{ atm}$ .

The sensitivity of propellant is  $0.005$  at  $20 \text{ atm}$  and  $0.0095$  at  $70 \text{ atm}$ . The army has complained that if the same missile is used in Florida ( $T = 40^\circ\text{C}$ ), the missile explodes.

check for the stability of the propellant  $n > 1$ ? at this temperature.

(P-3)



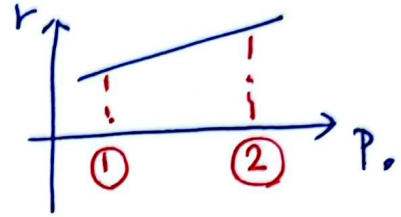
Also, at what maximum temperature can this missile operate stably?

**Solution:**

At Alaska,  $T_{i_1} = -10^\circ\text{C}$

@ 20 atm  $r = 3 \frac{\text{mm}}{\text{s}}$

@ 70 atm  $r = 8 \frac{\text{mm}}{\text{s}}$



$$n = \frac{\ln\left(\frac{r_2}{r_1}\right)}{\ln\left(\frac{P_{02}}{P_{01}}\right)} = 0.783 < 1$$

stable in Alaska

At FL, we should find "n" from  $\pi_r$

$$\pi_r = \frac{2}{r_1 + r_2} \frac{r_2 - r_1}{T_2 - T_1}$$

At 20 atm,  $\pi_r = 0.005$ ,  $\left\{ \begin{array}{l} r \text{ @ Alaska is } 3 \frac{\text{mm}}{\text{s}} \\ r_1 : r \text{ @ Alaska} \\ r_2 : r \text{ @ FL} \\ T_1 = -10^\circ\text{C} \\ T_2 = 40^\circ\text{C} \end{array} \right.$

$$0.005 = \frac{2}{r_2 + 3} \times \frac{r_2 - 3}{40 - (-10)}$$

Solve for  $r_2$ ,  $r_2 = 3.857 \frac{\text{mm}}{\text{s}}$

At 70 atm  $\left\{ \begin{array}{l} r \text{ @ Alaska} = r'_1 = 8 \frac{\text{mm}}{\text{s}} \\ r \text{ @ FL} = r'_2 = ? \\ T_1 = -10^\circ\text{C} \quad T_2 = 40^\circ\text{C} \end{array} \right.$

$$\frac{\pi_r}{0.0095} = \frac{2}{r'_2 + 8} \frac{r'_2 - 8}{40 - (-10)} \Rightarrow r'_2 = 12.983$$

(P-4)